Modular Forms

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Lecture 1

1 Introduction

Definition 1.1

Define the upper half plane

$$\mathfrak{H} = \{\tau \in \mathbb{C} \mid \mathrm{im}(\tau) > 0\}$$

and the positive determinant group

$$\mathrm{GL}_2(\mathbb{R})^+ = \{g \in \mathrm{GL}_2(\mathbb{R}) \mid \det(g) > 0\}$$

and

$$\Gamma(1) = \mathrm{SL}_2(\mathbb{Z})$$

Lemma 1.2. $\mathrm{GL}_2(\mathbb{R})^+$ acts transitively on $\mathfrak{H}$ via Möbius transformations.

Proof. Suppose $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})^+$, and $\tau \in \mathfrak{H}$. Then

$$\mathrm{Im}(g\tau) = \frac{1}{2i}\left(\frac{a\tau + b}{c\tau + d} - \frac{a\overline{\tau} + b}{c\overline{\tau} + d}\right) = \frac{1}{2i}\frac{(ad - bc)(\tau - \overline{\tau})}{|c\tau + d|^2} = \frac{\deg(g)\,\mathrm{Im}(\tau)}{|c\tau + d|^2} > 0$$

For transitivity, for $x + iy \in \mathfrak{H}$,

$$\begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix} \cdot i = x + iy$$

□

Definition 1.3

Let $g \in \mathrm{GL}_2(\mathbb{R})^+$, $\tau \in \mathfrak{H}$, define $j(g, \tau) = c\tau + d$, where $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This is called the *modular cocycle*.

If k is an integer, $f : \mathfrak{H} \to \mathbb{C}$ is any function, then

$$\begin{aligned} f|_k[g] : \mathfrak{h} &\to \mathbb{C} \\ \tau &\mapsto \det(g)^{k-1} f(g\tau) j(g, \tau) \end{aligned}$$

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is called the *weight k action of g on f*.

Lemma 1.4. This is a *right* action of $\mathrm{GL}_2(\mathbb{R})^+$. That is,

$$f|_k[gh] = (f|_k[g])\,|_k[h]$$

Proof.

$$\begin{aligned} f|_k[g]|_k[h](\tau) &= \det(h)^{k-1} f|_k[g](h\tau) j(h,\tau)^{-k} = \det(h)^{k-1}\det(g)^{k-1} f(gh\tau) j(g,h\tau)^{-k} j(h,\tau)^{-k} \\ &\overset{?}{=} \det(gh)^{k-1} f(gh\tau) j(gh,\tau)^{-k} \end{aligned}$$

Therefore, suffices to show that

$$j(gh,\tau) = j(g,h\tau)j(h,\tau)$$

Note that if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $g\begin{pmatrix}\tau\\1\end{pmatrix} = \begin{pmatrix} a\tau+b \\ c\tau+d\end{pmatrix} = j(g,\tau)\begin{pmatrix} g\tau \\ 1\end{pmatrix}$, This means that,

$$j(gh,\tau)\begin{pmatrix} gh\tau \\ 1\end{pmatrix} = gh\begin{pmatrix}\tau\\1\end{pmatrix} = g\left(j(h,\tau)\begin{pmatrix} h\tau\\1\end{pmatrix}\right) = j(g,h\tau)j(h,\tau)\begin{pmatrix} gh\tau\\1\end{pmatrix}$$

□

Formulae:
For $g, h \in \mathrm{GL}_2(\mathbb{R})^+$, $\tau \in \mathfrak{h}$,

$$\mathrm{Im}(g\tau) = \frac{\det(g)\,\mathrm{Im}(\tau)}{|j(g,\tau)|^2}$$

and

$$j(gh,\tau) = j(g,h\tau)j(h,\tau)$$

Definition 1.5

Let $k \in \mathbb{Z}$, $\Gamma \leq \Gamma(1)$ a finite index subgroup. A *weakly modular function of weight k and level Γ* is a meromorphic function $f : \mathfrak{H} \to \mathbb{C}$, which is invariant under the weight k action of Γ. That is, such that for all $\gamma \in \Gamma$,

$$f|_k[\gamma] = f$$

We will define a modular form (when $\Gamma = \Gamma(1)$) next time, but they are weakly modular functions, which are holomorphic in $\mathfrak{H}$ and at ∞.

In fact, modular forms of fixed weight and level live in finite dimensional $\mathbb{C}$-vector spaces $M_k(\Gamma)$, which are the main objects of study in this course.

Why do we study modular forms?

1. They are related to the theory of elliptic functions. Let $E/\mathbb{C}$ be an elliptic curve. Let ω be a non-zero holomorphic 1-form. Then there exists a unique lattice $\Lambda \leq \mathbb{C}$, and an isomorphism of Riemann surfaces, $\phi : \mathbb{C}/\Lambda \to E$, such that $\phi^*(\omega) = dz$. We can show that E is isomorphic to the elliptic curve $y^2 = 4x^3 - 60G_4(\Lambda)x - 140G_6(\Lambda)$, where for $k \in \mathbb{Z}$,

$$G_k(\Lambda) = \sum_{\lambda \in \Lambda\setminus 0} \lambda^{-k}$$

which converges for $\lambda > 2$. If $\tau \in \mathfrak{H}$, then we have an associated lattice $\Lambda_\tau = \mathbb{Z} \oplus \mathbb{Z}_\tau$, and the function

$$G_k(\tau) = G_k(\Lambda_\tau)$$

is a modular form of weight k and level $\Gamma(1)$. This is called the *Eisenstein series*. Moreover, $\mathfrak{H}/\mathrm{SL}_2(\mathbb{Z})$ can be identified (as a set) with the set of isomorphism classes of elliptic curves over $\mathbb{C}$.

2. Modular forms f have Fourier expansions
$$\sum_{n\in\mathbb{Z}} a_n q^n$$
where $a_n \in \mathbb{C}$, and often serve as generating functions for arithmetically interesting sequences a_n. One example is
$$\vartheta(q) = \sum_n e^{\pi i n^2 \tau}$$
If $k \in 2\mathbb{N}$, then ϑ^{2k} is a modular form, and on the other hand,
$$\vartheta^{2k} = \sum_{n\in\mathbb{Z}} r_k(n) e^{\pi i n \tau}$$
where $r_k(n)$ is the number of ways to write n as a sum of k squares. By expressing ϑ^{2k} in terms of ther modular forms, we can prove formulae such as
$$r_4(n) = 8 \sum_{d|n, 4\not|d} d$$

3. The Riemann ζ function is an important object in number theory. Properties include
   (a) The Euler product
   $$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$
   (b) A meromorphic extension to $\mathbb{C}$,
   (c) A functional equation relating $\zeta(s)$ and $\zeta(1-s)$.

   A *Dirichlet L-series* is a function of the form
   $$\sum_{n\geq 1} a_n n^{-s}$$
   which has similar properties is called an L-function. Modular forms can be used to construct interesting examples of L-functions. Take $M_k(\Gamma)$ and decompose them under the action of Hecke operators. In particular, if $\Gamma = \Gamma(1)$, we get a decomposition into lines, called Hecke eigenforms.

4. Connections to the Langland programme, which predicts (among other things) a relation between modular forms and other objects in arithmetic geometry. A special case of this is the *Modularity conjecture*, which says that there is a bijection between elliptic curves over $\mathbb{Q}$ (up to isogeny) and the set of Hecke eigenforms of weight 2. This implies Fermat's last theorem. The bijection is formulated in the language of Hecke operators and L-functions.

   Homework: Handout on Moodle called "Reminder of Complex Analysis".

Lecture 2

2 Modular forms on $\Gamma(1)$

Recall a *meromorphic function* on $U \subseteq \mathbb{C}$ is a closed subset $A \subseteq U$ and a holomorphic function $f : U \setminus A \to \mathbb{C}$, such that for every $a \in A$, there exists $\delta > 0$, such that
$$D_*(a, \delta) \subseteq U \setminus A$$
and there exists an integer $n \geq \mathbb{Z}$ such that
$$(z-a)^n f(z)$$
defines a holomorphic function on $D(a, \delta)$. Such an $a \in A$ is called a *pole* of f. f then has a *Laurent expansion*
$$\sum_{m\in\mathbb{Z}} a_m (z-a)^m$$
which is absolutely convergent on $D(a, \delta)$.

Lemma 2.1. Let f be a weakly modular function of weight k and level $\Gamma(1)$. Then there exists a unique meromorphic function $\tilde{f}$ on $D_*(0,1)$ such that

$$f(\tau)\tilde{f}(e^{2\pi i\tau})$$

Proof. By assumption, f is meromorphic on $\mathfrak{H}$, let $\gamma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma(1)$. Then

$$f|_k[\gamma](\tau) = f(\gamma\tau) = f(\tau)$$

as f is invariant under the weight k action of γ. But

$$f(\gamma\tau) = f(\tau + 1)$$

Existence: Locally, let $a \in D_*(0,1), \delta > 0$ be such that $D(a,\delta) \subseteq D_*(0,1)$. Define $\tilde{f}$ in the disc by

$$\tilde{f}(q) = f\left(\frac{1}{2\pi i}\log(q)\right)$$

where log is any branch of logarithm defined in $D(a,\delta)$. This is independent of the choice of branch of logarithm, since any two branches differ by $2\pi i$, and $f(\tau) = f(\tau+1)$. Therefore, this defines $\tilde{f}$ on $D_*(0,1)$.

Uniqueness: Since the map $\tau \mapsto e^{2\pi i\tau}$ is surjective, $\tilde{f}$ is unique. □

Suppose $\tilde{f}$ extends to a meromorphic function on $D(0,1)$, then there exists $\delta > 0$ such that $\tilde{f}$ has a Laurent expansion

$$\tilde{f}(q) = \sum_{n\in\mathbb{Z}} a_n q^n$$

which is absolutely convergent on $D_*(0,\delta)$. In particular, in the region where $\mathrm{Im}(\tau) > \frac{1}{2\pi}\log(\delta)$, we have

$$f(\tau) = \sum_{n} a_n q^n$$

where $q = e^{2\pi i\tau}$. We call this the *q-expansion of a weakly modular function f*.

Definition 2.2

Let f be a weakly modular function of weight k and level $\Gamma(1)$. We say that f is *meromorphic at ∞* if $\tilde{f}$ extends to a meromorphic function on $D(0,1)$.

We say that f is *holomorphic at ∞* if $\tilde{f}$ extends, and is holomorphic at $q = 0$. In this case, we define

$$f(\infty) = \tilde{f}(0) = \lim_{\mathrm{Im}(\tau)\to\infty} f(\tau)$$

We say that f *vanishes at ∞* if f is holomorphic at ∞, and $f(\infty) = 0$.

Definition 2.3 (modular form)

A *modular function* (of weight k and level $\Gamma(1)$) is a weakly modular function, which is meromorphic at ∞.
A *modular form* is a weakly modular function which is holomorphic in $\mathfrak{H}$, and holomorphic at ∞.

A *cuspidal modular form* is a modular form which vanishes at ∞.

Remark 2.4. We let

$$M_k(\Gamma(1))$$

be the set of modular forms of weight k and level $\Gamma(1)$, and

$$S_k(\Gamma(1)) \subseteq M_k(\Gamma(1))$$

be the set of cuspidal modular forms of weight k and level $\Gamma(1)$. These are $\mathbb{C}$-vector spaces. If k is odd, then they are zero. To see this, consider the matrix

$$\gamma = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \in \Gamma(1)$$

and the weight k-action is

$$f|_k[\gamma](\tau) = (-1)^k f(\tau) = f(\tau)$$

We will now consider even weights only.

If $k \in \mathbb{Z}$ is even, let

$$G_k(\tau) = \sum_{\lambda \in \Lambda_\tau \setminus 0} \lambda^{-k} = \sum_{(m,n) \in \mathbb{Z}^2 \setminus 0} (m\tau + n)^{-k}$$

where $\Lambda_\tau = \mathbb{Z}_\tau \oplus \mathbb{Z} \leq \mathbb{C}$.

If $\gamma \in \Gamma(1)$, then we formally have

$$G_k|_k[\gamma](\tau) = G_k(\gamma\tau) j(\gamma, \tau)^{-k} = \sum_{\lambda \in \Lambda_{\gamma\tau} \setminus 0} \lambda^{-k} j(\gamma, \tau)^{-k}$$

But if $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\Lambda_{\gamma\tau} = \mathbb{Z}\frac{a\tau + b}{c\tau + d} \oplus \mathbb{Z} = (c\tau + d)^{-1} \left(\mathbb{Z}(a\tau + b) \oplus \mathbb{Z}(c\tau + d)\right) = (c\tau + d)^{-1}\Lambda_\tau$$

Therefore,

$$G_k|_k[\gamma](\tau) = \sum_{\lambda \in (c\tau + d)^{-1}\Lambda_\tau \setminus 0} \lambda^{-k}(c\tau + d)^{-k} = \sum_{\lambda \in \Lambda_\tau \setminus 0} \lambda^{-k} = G_k(\tau)$$

This is justified only when the series defining $G_k(\tau)$ converges absolutely.

Proposition 2.5. Let $k > 2$ be an even integer. Then $G_k(\tau)$ converges absolutely, and defines a modular form of weight k and level $\Gamma(1)$, with

$$G_k(\infty) = 2\zeta(k)$$

where ζ is the Riemann ζ-function. We call G_k the *weight k Eisenstein series*.

Remark 2.6. We will see later that $M_2(\Gamma(1)) = 0$, so this is optimal.

Proof. We want to show absolute and local uniform convergence in $\mathfrak{H}$, since this shows that G_k is holomorphic.

Let $A \geq 2$, and define

$$\Omega_A = \left\{ \tau \in \mathfrak{H} \;\middle|\; \operatorname{Im}(\tau) \geq \frac{1}{A} \text{ and } \operatorname{Re}(\tau) \in [-A, A] \right\}$$

We will show uniform convergence in Ω_A. If $\tau \in \Omega_A$, $x \in \mathbb{R}$, then

$$|\tau + x| \geq \begin{cases} \frac{1}{A} & |x| \leq 2A \\ \frac{|x|}{2} & |x| \geq 2A \end{cases}$$

Therefore,

$$\tau + x \geq \max\left\{\frac{1}{A}, \frac{|x|}{2A^2}\right\} \geq \max\left\{\frac{1}{A^2}, \frac{|x|}{2A^2}\right\} \geq \frac{1}{2A^2}\max\{1, |x|\}$$

If $(m, n) \in \mathbb{Z}^2$, then we get that

$$|m\tau + n| \geq \frac{1}{2A^2}\max\{|m|, |n|\}$$

If $\tau \in \Omega_A$, then

$$\begin{aligned}
\sum_{(m,n) \in \mathbb{Z}^2 \setminus 0} |m\tau + n|^{-k} &\leq (2A^2)^k \sum_{(m,n) \in \mathbb{Z}^2 \setminus 0} \max\{|m|, |n|\}^{-k} \\
&= (2A^2)^k \sum_d d^{-k} \cdot \#\{(m, n) \in \mathbb{Z}^2 \setminus 0 \mid \max\{|m|, |n|\} = d\} \\
&= (2A^2)^{-k} \sum_d d^{1-k} \\
&= 8(2A^2)^k \zeta(k-1) < \infty
\end{aligned}$$

as $k > 2$. This shows uniform convergence in Ω_A by the Weierstrass M-test.

We now know that G_k is holomorphic in $\mathfrak{H}$, and invariant under the weight k action of $\Gamma(1)$. It remains to show to show G_k is holomorphic at ∞, with $G_k(\infty) = 2\zeta(k)$. It suffices to show that

$$\lim_{\mathrm{Im}(\tau)\to\infty} G_k(\tau) = 2\zeta(k)$$

By uniform convergence, we can exchange the limit and sum. But

$$\lim_{\mathrm{Im}(\tau)\to\infty} (m\tau + n)^{-k} = \begin{cases} 0 & m \neq 0 \\ n^{-k} & m = 0 \end{cases}$$

□

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